

SEMARANG AHMAD YANI, Indonesia

WMO: 968390

Lat: 6.9769S

Lon: 110.3778E

Elev: 13

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	72.6	73.7	59.0	74.6	89.0	61.7	82.3	87.0	22.3	85.4	20.0	85.2	2.9	120	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	14.6	94.6	74.3	93.1	75.2	91.8	75.8	80.8	87.9	80.1	87.3	79.8	87.0	11.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	150.4	85.1	78.3	147.1	84.6	77.7	143.8	84.0	44.5	88.0	43.9	87.8	43.5	87.5	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	15.9	14.1	DB	68.5	97.8	3.0	1.9	66.3	99.2	64.5	100.3	62.8	101.4	60.6	102.8	(4)
(5)			WB	63.4	83.9	2.5	1.5	61.6	85.0	60.2	85.9	58.8	86.8	57.0	87.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.0	81.5	81.3	82.3	83.4	84.0	83.4	82.8	83.1	84.1	84.5	83.5	82.1	(6)	
	DBStd	1.95	1.75	1.69	1.52	1.53	1.61	1.60	1.56	1.58	1.77	1.97	2.01	1.57	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	12050	975	877	1002	1002	1055	1002	1017	1027	1024	1071	1004	994	(10)	
	CDD65	6575	510	457	537	552	590	552	552	562	574	606	554	529	(11)	
	CDH74	73488	5092	4537	5807	6288	7061	6362	6193	6400	6653	7256	6300	5539	(12)	
	CDH80	28335	1505	1299	1940	2381	2963	2593	2519	2695	2904	3214	2517	1805	(13)	
(14) Wind	WSAvg	5.9	6.2	6.1	5.4	5.3	5.8	5.9	6.3	6.7	6.5	6.1	5.2	5.0	(14)	
(15) Precipitation	PrecAvg	68.8	15.4	8.8	8.6	6.7	4.8	2.9	1.9	0.8	1.9	3.5	5.5	8.0	(15)	
	PrecMax	102.8	34.2	19.8	18.3	14.8	13.0	6.3	7.2	4.4	11.8	10.3	19.2	21.6	(16)	
	PrecMin	34.7	6.9	2.8	2.9	1.0	0.7	0.0	0.0	0.0	0.0	0.0	1.5	3.8	(17)	
	PrecStd	22.2	9.3	4.8	4.5	4.2	3.4	2.4	2.0	1.3	3.5	3.1	4.6	4.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.7	89.9	90.9	92.6	93.3	93.1	92.8	93.6	96.5	96.9	95.3	91.1	(19)	
		MCWB	78.8	79.1	78.4	78.6	77.8	75.7	74.3	73.1	72.3	73.0	76.7	78.8	(20)	
	2%	DB	88.8	88.2	89.4	91.2	91.8	91.6	91.6	92.6	94.6	94.3	92.8	89.4	(21)	
		MCWB	78.8	78.8	79.0	78.6	77.6	75.9	74.3	73.3	72.6	74.9	77.6	78.9	(22)	
	5%	DB	87.6	87.3	88.1	89.7	91.0	90.7	90.6	91.4	92.6	92.1	90.9	88.1	(23)	
		MCWB	78.4	78.7	78.8	78.6	77.6	76.2	74.5	73.5	73.7	76.1	78.0	78.7	(24)	
	10%	DB	86.3	86.1	87.3	88.5	89.7	89.5	89.5	89.9	90.6	90.4	89.3	87.2	(25)	
		MCWB	78.1	78.3	78.6	78.7	77.7	76.5	74.8	74.1	74.9	76.8	78.2	78.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.5	80.5	81.0	81.4	81.4	80.7	79.7	79.6	79.6	80.1	80.9	80.7	(27)	
		MCDB	87.4	86.9	87.6	88.3	88.4	87.5	86.9	86.9	88.0	88.3	88.9	87.6	(28)	
	2%	WB	79.6	79.8	80.1	80.5	80.3	79.8	78.5	78.3	78.5	79.3	79.8	79.9	(29)	
		MCDB	86.3	86.1	86.8	87.5	87.8	86.9	86.1	86.2	86.7	87.8	87.9	86.8	(30)	
	5%	WB	79.0	79.2	79.5	79.9	79.8	79.1	77.8	77.5	77.8	78.8	79.1	79.2	(31)	
		MCDB	85.6	85.3	86.2	86.8	87.1	86.3	85.4	85.4	86.1	87.4	87.2	86.0	(32)	
	10%	WB	78.5	78.7	79.0	79.3	79.1	78.4	77.1	76.8	77.1	78.2	78.6	78.8	(33)	
		MCDB	84.8	84.7	85.5	86.1	86.4	85.7	84.8	84.7	85.5	86.6	86.6	85.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	9.8	10.6	11.6	12.5	13.4	14.8	15.7	16.0	14.6	12.8	10.7	(35)	
		MCDBR	11.5	11.2	11.6	12.8	13.6	14.6	15.9	16.9	18.0	16.5	15.1	11.8	(36)	
	5% WB	MCWBR	4.7	4.7	4.7	4.5	4.8	5.4	6.1	6.6	6.6	5.7	5.0	4.7	(37)	
		MCWBR	10.7	10.5	10.9	11.7	12.4	12.4	13.9	14.6	15.1	14.2	12.9	11.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.535	0.542	0.542	0.554	0.526	0.515	0.518	0.549	0.580	0.662	0.645	0.578	(40)	
		taud	2.090	2.074	2.073	2.026	2.093	2.114	2.094	1.991	1.913	1.728	1.778	1.952	(41)	
	Ebn at Noon		259	257	253	242	239	237	239	239	240	226	230	247	(42)	
		Edh at Noon	55	56	55	55	49	47	49	56	63	78	74	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1454	1470	1592	1626	1603	1541	1639	1830	1958	1866	1651	1462	(44)	
	RadStd		144	125	96	97	89	103	100	69	109	167	155	123	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.58	N/A	N/A	+0.70	+0.63	+0.85	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.58	N/A	N/A	+0.70	+0.63	+0.85	N/A	N/A	N/A

Nomenclature: See separate page